TG-LiteNet: A Topology-Aware, Domain-Generalizable and Lightweight Framework for Retinal Vessel Segmentation in Resource-Constrained Screening Settings

CSD361 – Introduction to Machine Learning, Final Project Report

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***Abstract*—Retinal vessel segmentation underpins screening for diabetic retinopathy, glaucoma and hypertensive retinopathy, but a model deployed in a rural screening camp must (i) keep thin vessels *connected*, (ii) work on cameras it was never trained on, and (iii) run on modest hardware. We study the three-way trade-off with *TG-LiteNet*, a compact depthwise-separable encoder–decoder trained on a single source dataset (DRIVE) with a soft-clDice topology loss, an auxiliary skeleton head, domain-generalisation (DG) augmentation with a consistency loss, and Monte-Carlo-dropout uncertainty. A staged ablation with 5 seeds per stage, a $2\times2\times2$ factorial design and capacity/loss-variant sweeps are evaluated on three held-out datasets (STARE, CHASE_DB1, HRF) using Dice, clDice, Betti-number error, parameters, FLOPs, latency and uncertainty–error correlation. TG-LiteNet has 0.21 M parameters and 5.5 GFLOPs (U-Net: 31.04 M, 384.7 GFLOPs; 147$\times$ and 70$\times$ fewer) and reaches an unseen-domain Dice of 0.756 and clDice of 0.833, versus 0.762 and 0.835 for the U-Net baseline. Topology supervision more than halves the raw unseen Betti-0 error of the U-Net (75.0 to 31.7) and adds 0.007 clDice, but removing small components, with the size chosen on DRIVE validation, brings every model to about 5, so the connectivity gain is not specific to the loss. DG augmentation adds about 0.01 unseen Dice. Plain compact U-Nets match the depthwise-separable backbone in accuracy and are as fast or faster on a CPU, and removing the skeleton head or the consistency loss costs no Dice or clDice. MC-dropout uncertainty ranks pixel errors moderately ($\rho_{px} \approx 0.31$–0.34), but image-level rejection is not reliably better than random. All numbers are produced by the released code; no target-domain image or label is used for training, tuning or model selection.**

***Keywords*—retinal vessel segmentation, topology-aware loss, domain generalization, lightweight networks, uncertainty estimation, DRIVE**

I. Introduction

Segmenting the retinal vascular tree from fundus photographs supports the measurement of vessel calibre, tortuosity and branching used in screening for diabetic retinopathy, glaucoma and hypertensive retinopathy [1]. Since U-Net [2] such networks are trained and tested on small public datasets, typically DRIVE [3], STARE [4], CHASE_DB1 [5] and HRF [6]. Overlap scores hide topological errors: a single severed centreline barely changes Dice but changes the connectivity that downstream biomarkers rely on [7]–[9].

Motivation. A model that is useful in a low-resource screening programme must satisfy three constraints *simultaneously*: it must preserve vessel *topology*; it must tolerate camera and population *shift*, ideally without target labels, because performance falls when a model meets a new database [12], [13]; and it must be small and fast enough for *edge* hardware [14], as in screening camps with limited compute and connectivity. Recent papers each address part of this, so our contribution is not a simple "combination of the three".

Problem statement. Let f_θ be trained on a labelled source domain $\mathcal{S}$ (DRIVE) and evaluated, with no access to target images or labels, on unseen domains $\mathcal{T} = \{\text{STARE, CHASE_DB1, HRF}\}$. For a metric m the *target gap* is $\Delta_m(\mathcal{T}) = m(\mathcal{S}) - m(\mathcal{T})$. The goal is to keep Δ_m small for both overlap and topology metrics under a compact parameter and latency budget, and to flag unreliable predictions with an uncertainty estimate.

Objectives. (O1) Build a reproducible *source-only* pipeline for topology-aware vessel segmentation; (O2) quantify what each component (topology loss, DG augmentation, compact backbone, uncertainty) contributes to accuracy, topology, cross-domain gap and cost in one controlled ablation; (O3) test whether MC-dropout uncertainty can triage unreliable predictions, especially on unseen domains.

Approach and contributions. We propose *TG-LiteNet*, a depthwise-separable encoder–decoder with a mask head and an auxiliary skeleton head. Our contributions are: (1) a staged ablation and a full $2\times2\times2$ factorial study (backbone $\times$ topology loss $\times$ DG augmentation) measured with repeated seeds and confidence intervals; (2) a unified evaluation that reports Dice/IoU, clDice, Betti-0/1 error, the source-to-

target gap, total parameters, FLOPs, CPU/GPU latency and uncertainty quality *together*; (3) an uncertainty-based triage analysis at pixel and image level, separately for the source and each unseen domain; (4) controls that test whether the gains survive simple alternatives (small-component removal, validation-tuned thresholds, plain compact U-Nets, removing single ingredients); and (5) open code that regenerates every number in this paper. We do not claim a new loss or a new architecture family, and we make no state-of-the-art claim.

II. Related Work

Base paper: clDice. Shit et al. [7] compare the skeleton of a prediction with the ground-truth mask (topology precision) and the skeleton of the ground truth with the predicted mask (topology sensitivity); their harmonic mean is clDice. They prove that maximising clDice preserves topology up to homotopy equivalence for binary tubular structures and give a differentiable *soft-clDice* based on iterated soft morphological opening. On five datasets, including DRIVE, clDice-trained networks reduce Betti errors at a small training-time overhead. cbDice [8] observes that clDice favours large vessels ("diameter imbalance") and adds radius and boundary information; Skeleton Recall Loss [9] computes the skeleton once, offline, with cheap CPU operations; earlier work used persistent homology [10]. None of these is evaluated for source-only robustness under dataset shift.

Domain generalisation and adaptation. "Cross-dataset" covers four protocols: source-only transfer, multi-source DG, test-time adaptation (TTA) and fine-tuning with target labels. DGSSA [15] synthesises vessel trees by space colonisation and adds style augmentation (multi-source); WaveRNet [16] separates illumination-robust low frequencies with a SAM backbone; Hu et al. [17] and GraphSeg [18] use a Hessian vector field and a deformable graph prior. A benchmark [13] finds that, with enough training data, a plain U-Net generalises well.

Lightweight networks. Galdran et al. [12] show that U-Nets with ∼34K and W-Nets with ∼68K parameters are competitive; DSAE-Net [19] (78K parameters, skeleton-distance loss) and LightVesselNet [14] (75K parameters) report cross-dataset Dice but no connectivity metrics or uncertainty; a reverse-attention U-Net [20] targets embedded use.

Topology under shift; uncertainty. TFFM [21] (source-only transfer), TopoTTA [22], TopoLoRA-SAM [23] and TAPDecoderFT [11] report clDice/Betti after zero-shot, test-time or fine-tuned transfer, but with large or unspecified models and no latency. Topology-aware uncertainty [24] scores whole structures; MC dropout [25] is a cheap pixel-level alternative whose quality degrades under dataset shift [26].

Research gap. Table I summarises what the closest works report. Among the works we reviewed, none evaluates a compact *source-only* model for topology, total deployment cost and uncertainty quality in one controlled protocol with repeated runs. This project fills exactly that gap, as a trade-off *study* rather than a new method.

TABLE I
EVIDENCE REPORTED BY THE CLOSEST REVIEWED WORKS (✓: REPORTED, ∼: PARTIAL/INDIRECT, –: NOT REPORTED). PROTOCOL: SO = SOURCE-ONLY, MS = MULTI-SOURCE DG, TTA, FT = FINE-TUNING ON LABELLED TARGET DATA.

Work	Protocol	Topo. loss	Topo. metrics	Total cost	Uncert.
clDice / SRL [7], [9]	none	✓	✓	∼	–
DGSSA, WaveRNet [15], [16]	MS	–	–	–	–
Minimal., DSAE, LightVessel [12], [14], [19]	train 1, test many	∼	–	✓	–
TFFM [21]	SO	✓	✓	–	–
TopoTTA [22]	TTA	∼	✓	n/v	–
TopoLoRA-SAM, TAPDecoderFT [11], [23]	FT	✓	✓	∼	–
Topology-aware unc. [24]	–	–	∼	n/v	✓
TG-LiteNet (this work)	SO	✓	✓	✓	✓

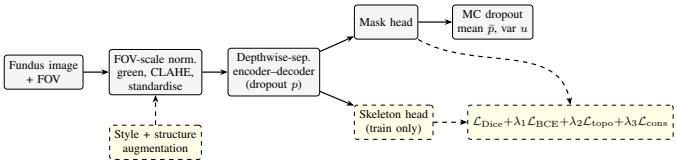


Fig. 1. TG-LiteNet pipeline. Dashed/yellow elements are used only during training; one fixed model is evaluated on every target.

III. Problem Formulation and Research Questions

Given a fundus image I with field-of-view (FOV) mask F, predict a binary vessel mask $M \in \{0,1\}^{H\times W}$. Models are trained on DRIVE only; STARE, CHASE_DB1 and HRF are never used for training, tuning or model selection. We ask:

- **RQ1 (topology).** Does topology supervision (soft-clDice + skeleton head) improve clDice and Betti error relative to Dice+BCE without an unacceptable loss of overlap?
- **RQ2 (generalisation).** Does lightweight style-plus-structure augmentation with a consistency loss reduce Δ_m from DRIVE to the unseen targets?
- **RQ3 (trade-off).** What accuracy and topology cost does a $\leq$2M-parameter budget incur relative to the savings in parameters, FLOPs and latency?
- **RQ4 (uncertainty).** Does MC-dropout uncertainty rank high-error cases and support selective rejection, more so on unseen domains than on the source domain?

IV. Proposed Methodology

A. Overview

Fig. 1 shows the pipeline. A fundus image is rescaled so that its FOV has a fixed width, reduced to the CLAHE-enhanced, standardised green channel, and passed through an encoder–decoder with a mask head (and, during training only, a skeleton head). The loss combines region, pixel, topology and consistency terms. At test time, T stochastic forward passes give a mean probability map and a variance map used as uncertainty.

B. Preprocessing and scale normalisation

Cameras differ strongly in resolution (HRF images are ∼6× wider than DRIVE's), so the pixel width of vessels, to which a convolutional model is very sensitive, changes with the dataset. Every image is rescaled so that its FOV is W_F=527 px wide (DRIVE's median), which uses neither labels nor any statistic of the target *dataset*. We then take the green channel (best

vessel contrast), replace the outside of the FOV by the FOV mean, apply CLAHE [27] with a fixed tile size in pixels, standardise inside the FOV, and set the outside to zero. The same deterministic function is used in training and testing.

C. Architectures

Baseline U-Net [2]: four 2×2 max-pool stages, double 3×3 conv+BN+ReLU blocks, transposed-convolution up-sampling, base width 64 (≈31.04 M parameters).

TG-LiteNet backbone. Every 3×3 convolution is replaced by a depthwise 3×3 followed by a pointwise 1×1 convolution (BN, ReLU6), as in MobileNets [28]. A block has two such layers and a 1×1 residual shortcut. Encoder widths are $(w, 2w, 4w, 8w, 8w)$ with w=16; the decoder uses bilinear up-sampling, skip concatenation and one separable block per level. A 1×1 mask head outputs vessel logits; an auxiliary 1×1 skeleton head (training only, no inference cost) predicts the 1-px-dilated skeleton of the label. Spatial dropout (p=0.1) after the bottleneck and each decoder block is used only in stage E. The width w is the single capacity knob; Section VII sweeps it.

D. Objective

For a prediction $P = \sigma(z) \in [0,1]^{H\times W}$, label G and FOV F, the total loss is

$$\mathcal{L} = \mathcal{L}_{\text{Dice}} + \lambda_1 \mathcal{L}_{\text{BCE}} + \lambda_2 \mathcal{L}_{\text{topo}} + \lambda_3 \mathcal{L}_{\text{cons}}, \qquad (1)$$

with all terms computed inside F. $\mathcal{L}_{\text{Dice}}$ and $\mathcal{L}_{\text{BCE}}$ are the usual soft-Dice and binary cross-entropy.

Topology term. With $S(\cdot)$ the (soft) morphological skeleton, topology precision and sensitivity are $T_{\text{prec}} = \frac{\sum S(P)\odot G}{\sum S(P)}$, $T_{\text{sens}} = \frac{\sum S(G)\odot P}{\sum S(G)}$, and

$$\mathcal{L}_{\text{clDice}} = 1 - \frac{2\,T_{\text{prec}} T_{\text{sens}}}{T_{\text{prec}} + T_{\text{sens}}}. \qquad (2)$$

$S(P)$ is the differentiable skeleton obtained by iterating soft erosion/opening (10 iterations, enough for vessels up to 10 px radius at our scale); $S(G)$ is the exact skeleton of the (augmented) label, computed in the data loader. The skeleton head adds $\mathcal{L}_{\text{skel}} = \text{Dice} + \text{BCE}$ against the dilated skeleton, so $\mathcal{L}_{\text{topo}} = \mathcal{L}_{\text{clDice}} + \gamma \mathcal{L}_{\text{skel}}$. As supplementary variants we implemented the binary Skeleton Recall loss [9], $\mathcal{L}_{\text{SR}} = 1 - \sum P \odot \mathcal{T}(S(G)) / \sum \mathcal{T}(S(G))$ with $\mathcal{T}$ a 3×3 dilation, and their sum. We also re-implemented the radius- and boundary-aware cbDice [8] (the cl-MIN-Dice variant) from its equations: with D the distance-to-boundary maps and r the skeleton radii, $T_{\text{prec}} = \sum_{S_P} \frac{D_L}{r_P} \big/ \big(|S_P \setminus S_L| + \sum_{S_P \cap S_L} \frac{r_L}{r_P}\big)$ and symmetrically T_{sens}, combined as in Eq. (2). It coincides with clDice for aligned centrelines but gives only partial credit to a predicted centreline that is displaced inside the vessel, and is meant to be paired with Dice. The label's skeleton and distance map are exact; for the prediction we use the soft skeleton and a differentiable soft-erosion distance map (so it may differ numerically from the authors' code). cbDice is evaluated as a metric for every model and tested as a loss variant (Sec. VII).

Algorithm 1 Training one TG-LiteNet configuration (one seed)

```
Pre-compute standardised green+CLAHE inputs of the 16 training images (DRIVE, scale-normalised)
for iteration = 1, ..., N do
    sample B patches: random affine crop of x, label G, FOV F, exact skeleton S(G)
    if DG enabled then
        x̃ ← style(x) with the same crop/elastic warp; local intensity dropout
    end if
    forward (x [and x̃]) → mask logits [and skeleton logits]
    L ← Eq. (1) (terms enabled by the stage)
    AdamW step (cosine schedule, gradient clipping)
    if iteration mod v = 0 then
        evaluate Dice on the 4 DRIVE validation images; keep the best checkpoint
    end if
end for
test: evaluate the checkpoint on DRIVE-test, STARE, CHASE_DB1, HRF (fixed threshold 0.5)
```

Consistency term (DG). For each patch we build a perturbed view $\tilde{x}$ (below) that is pixel-aligned with the clean view x: $\mathcal{L}_{\text{cons}} = \frac{1}{|F|}\sum_F \big(\sigma(f(\tilde{x})) - \text{sg}[\sigma(f(x))]\big)^2$, where sg stops gradients. Both views are also supervised.

E. Domain-generalisation augmentation

All stages use the same standard geometric augmentation (random crop at a FOV pixel, flips, free rotation, isotropic scaling in $[0.8, 1.25]$). Stage C onwards additionally applies, *only at training time*: **style** — random gain/gamma/contrast/brightness, smooth multiplicative illumination field, Gaussian blur, noise and JPEG compression of the green channel before CLAHE; and **structure** — an elastic deformation applied identically to both views and the label, and local intensity dropout (soft ellipses with attenuated contrast, view 2 only). Because CLAHE and standardisation follow the colour/gamma/contrast jitter, those three perturbations are largely neutralised by design (the same pipeline runs at test time); the effective style perturbations are the illumination field, blur, noise and JPEG. A DG stage processes two views per patch (clean and perturbed), i.e. twice the forward passes of stages A/B for the same number of iterations and patches; the elastic warp is capped so that it never folds (positive Jacobian). This is deliberately much simpler than synthesising vessel trees [15].

F. Uncertainty and triage

With dropout kept active (BatchNorm in evaluation mode), T passes give $p_t = \sigma(f_t(x))$, the mean $\bar{p} = \frac{1}{T}\sum_t p_t$ and the variance $u = \frac{1}{T}\sum_t p_t^2 - \bar{p}^2$ [25]. We compare u with the predictive entropy $H(\bar{p})$, which is also available for deterministic models. For triage, an image is scored by its mean u inside F; the images (or pixels) with the largest score are rejected first.

V. Experimental Setup

A. Datasets, preprocessing and exploratory analysis

Table II lists the datasets. DRIVE provides 20 training and 20 test images; we split the training images once (seed 2026)

TABLE II

DATASETS (ALL STATISTICS MEASURED ON THE FILES USED IN OUR EXPERIMENTS). FOV = FIELD-OF-VIEW WIDTH, USED TO NORMALISE THE SCALE OF EVERY IMAGE; CALIBRE = MEAN VESSEL WIDTH (2× DISTANCE TRANSFORM ON THE LABEL SKELETON) AT NATIVE RESOLUTION AND AFTER NORMALISATION TO A 527 PX FOV; G = GREEN-CHANNEL MEAN±STD INSIDE THE FOV (0–255). SOURCE = TRAIN (16) / VALIDATION (4) / TEST (20 OFFICIAL TEST IMAGES); TARGETS ARE TEST-ONLY.

Dataset	#	Size (px)	Vessel %	FOV (px)	Calibre (px)	G	Role
DRIVE [3]	40	565×584	12.9	527	3.5→3.5	97±17	source
STARE [4]	20	700×605	10.5	638	4.3→3.5	109±24	target
CHASE_DB1 [5]	28	999×960	10.3	899	6.1→3.6	62±26	target
HRF [6]	45	3504×2336	9.1	3262	8.8→1.4	60±17	target

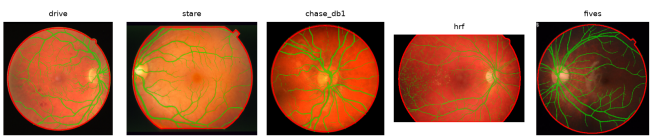


Fig. 2. Exploratory view of the datasets: vessel labels (green) and FOV outlines (red). Camera, colour balance, field of view and resolution differ markedly between datasets, which is the domain shift studied here.

into 16 for training and 4 for validation (model selection and, if used, hyper-parameter selection), and use the 20 test images as the in-domain test set. STARE (20), CHASE_DB1 (28) and HRF (45) are used only for testing. Ground truth is the first human observer. HRF and (when available) official DRIVE masks provide the FOV; for DRIVE (mirror), STARE and CHASE_DB1 the FOV is estimated by thresholding, keeping the largest component and eroding it by 1% (checked visually, Fig. 2). The vessel fraction is 9–13% in every dataset, i.e. about 1:8 class imbalance; there are no missing values or categorical attributes in this task. The exploratory statistics of Table II quantify the domain shift: the mean green-channel intensity ranges from ≈60 (CHASE_DB1, HRF) to ≈110 (STARE), and while the vessel calibre after FOV normalisation is ≈3.5 px for DRIVE, STARE and CHASE_DB1, it is only ≈1.4 px for HRF: HRF's reference annotations are thin relative to its field of view, so HRF measures an annotation-style shift in addition to the camera shift (a no-rescale evaluation is in Table VII). DRIVE and CHASE_DB1 were obtained from public mirrors (see `README.md`), STARE and HRF from the official sites; sources and checksums are recorded in `data/MANIFEST.json`.

B. Implementation and procedure

Everything is implemented in Python with PyTorch [29] (2.11.0+cu128), OpenCV, scikit-image [30] (Zhang–Suen thinning [31] for skeletons, connected components, Euler number) and scikit-learn. Training and efficiency measurements ran on free Kaggle notebooks with a Tesla T4 GPU and a 4-thread x86_64 CPU. All stages share the optimiser, schedule and data pipeline (Table III); stages differ only in the switches of Table IV. We train each stage A–E with 5 seeds and each additional configuration (factorial cells, loss variants, widths, controls) with 3 seeds, and report mean±std; differences between configurations are given with 95% Welch intervals (computed on per-seed unseen averages; all intervals are in `results/tables/stage_deltas.csv` and `factorial_deltas.csv`).

TABLE III

TRAINING CONFIGURATION SHARED BY ALL STAGES (GENERATED FROM `CONFIGS/BASE.YAML`).

Item	Value
Optimiser	AdamW, lr 0.001, wd 0.0001
Schedule	200 warm-up it., cosine to 1%
Iterations / batch / patch	5000 / 8 / 256^2 px
Loss weights	λ_1=1, λ_2=0.5, λ_3=0.5; skel. head 0.5
Soft-skeleton iterations	10
CLAHE	clip 2, 64 px tiles
Scale normalisation	FOV width = 527 px
Lite width w / U-Net base	16 / 64
MC dropout (stage E)	p=0.1, T=20 passes
Threshold	0.5 (fixed; never tuned on targets)

TABLE IV

STAGED ABLATION: EACH ROW ADDS EXACTLY ONE COMPONENT.

Stage	Topology loss	DG aug.+cons.	Compact backbone	MC dropout
A: U-Net baseline	–	–	–	–
B: + topology	✓	–	–	–
C: + DG	✓	✓	–	–
D: + compact backbone	✓	✓	✓	–
E: TG-LiteNet (full)	✓	✓	✓	✓

C. Evaluation metrics

All metrics are computed inside the FOV on the binarised prediction (threshold 0.5, fixed before seeing any target) against the *native-resolution* ground truth: the model runs at the normalised scale and its probability map is resampled (bilinear) to the label resolution, because down-scaling thin labels (HRF contains 1–2 px lines) fragments them and inflates Betti-0. Histogram-based AUC (4096 bins) is used for images above 2 MP (HRF). **Accuracy:** Dice, IoU, sensitivity, specificity, accuracy and AUC. **Topology:** clDice (computed with exact skeletons), cbDice (radius- and boundary-aware clDice variant, Sec. IV-D) and the Betti-0 / Betti-1 errors, i.e. the absolute difference in the number of connected components (β_0, 8-connectivity) and loops ($\beta_1 = \beta_0 - \chi$ from the Euler number). **Generalisation:** $\Delta_m = m(\text{DRIVE}) - m(\text{target})$; "unseen" is the macro average over the three targets. **Cost:** total parameters, FLOPs (2×MACs, PyTorch FLOP counter) and median latency for one 512×512 image (CPU with 1 and 4 threads, GPU), plus peak GPU memory. **Uncertainty:** per-image Spearman correlation between pixel uncertainty and misclassification (ρ_{px}), Spearman correlation between image-mean uncertainty and 1−Dice (ρ_{img}), pixel- and image-level risk–coverage curves and the area under the image-level curve (AURC) with random and oracle references.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Main results (staged ablation)

Table V gives the staged ablation; per-dataset values are in `results/tables/summary.csv`. Differences below are means over seeds with 95% Welch half-widths in brackets (training randomness). We also compare models on the

TABLE V

STAGED ABLATION (MEAN±STD OVER SEEDS). SOURCE = DRIVE TEST; UNSEEN = MACRO AVERAGE OVER STARE, CHASE_DB1 AND HRF (NEVER USED FOR TRAINING OR MODEL SELECTION); ΔDICE = DICE(DRIVE)−DICE(UNSEEN). COST: ONE 512×512 IMAGE, BATCH 1, CPU = 4 THREADS, KAGGLE MACHINE; LATENCY IS MACHINE-DEPENDENT; E (MC) NEEDS T FORWARD PASSES. BEST PER COLUMN IN BOLD.

	Source (DRIVE)		Unseen domains							Latency (ms)	
Stage	Dice	clDice	Dice	clDice	cbDice	β_0 err	ΔDice	Params (M)	GFLOPs	CPU	GPU
A	**0.830**±0.001	0.830±0.003	**0.762**±0.006	0.835±0.002	**0.732**±0.005	74.99±14.36	0.068±0.006	31.04	384.7	2915	91.1
B	0.827±0.001	0.845±0.001	0.753±0.004	0.843±0.003	0.728±0.003	31.69±2.27	0.074±0.004	31.04	384.7	2886	89.6
C	0.826±0.001	**0.849**±0.001	0.761±0.001	**0.847**±0.001	0.729±0.002	**27.93**±2.11	**0.065**±0.001	31.04	384.7	2770	89.3
D	0.822±0.001	0.838±0.003	0.757±0.003	0.832±0.002	0.722±0.004	44.71±6.49	0.065±0.003	0.21	5.5	405	12.2
E (det.)	0.822±0.001	0.839±0.003	0.755±0.005	0.833±0.002	0.720±0.005	41.95±2.84	0.066±0.005	0.21	5.5	429	12.2
E (MC)	0.822±0.001	0.838±0.003	0.756±0.004	0.833±0.002	0.721±0.005	42.66±2.98	0.066±0.004	0.21	5.5 × 20	429 × 20	12.2 × 20

same test images (paired bootstrap over the 93 unseen images, `paired_image_deltas.csv`), which captures test-set sampling; with 5 seeds (3 for supplementary runs) and no multiplicity correction both kinds of interval are descriptive.

Topology (A→B, RQ1). The topology loss lowers the raw unseen Betti-0 error from 75.0 to 31.7 ($-43.3\,[\pm 17.7]$, on every target) and raises unseen clDice by 0.007 (±0.004), at a cost of 0.009 unseen Dice (±0.008) and 0.003 on DRIVE (±0.002). On the 93 unseen images Dice is lower for 92% and clDice higher for 96%. It does not reduce loop errors: unseen Betti-1 moves only from 22.5 to 21.0 and is worse for the compact models (24.4). RQ1 therefore holds for centreline overlap and the raw fragment count, not for area overlap or loops, and the checks below show that the fragment count is mostly removable by simple cleanup.

Generalisation (B→C, RQ2). DG augmentation raises unseen Dice by 0.008 (±0.005; higher for 81% of the images) and shrinks the DRIVE-to-unseen gap from 0.074 to 0.065 (-0.009 ± 0.005). CHASE_DB1 sensitivity rises from 0.879 to 0.927, while unseen clDice (+0.004, ±0.004) and Betti-0 (-3.8, ± 3.2) change little. RQ2 holds for overlap, only marginally for topology.

Compact backbone (C→D, RQ3). Shrinking from 31 M to 0.21 M parameters costs 0.004 Dice on DRIVE (±0.001), 0.005 unseen Dice (±0.004; lower for 72% of the images) and clearly more topology: unseen clDice −0.014 (±0.003) and raw Betti-0 +16.8 (±7.9), mostly on HRF, whose thin annotations reward capacity (sensitivity 0.891 to 0.862). The gap is unchanged (0.065). Stage E (dropout, MC averaging) differs from D by at most 0.002 Dice (lower for 81% of the images) and not in clDice or Betti-0: dropout brings no benefit, only the uncertainty map. Overall (A→E), unseen Dice is 0.007 lower (±0.008), DRIVE Dice 0.008 lower (±0.002), unseen clDice 0.003 lower (±0.003) and raw unseen Betti-0 32 lower (±18).

Classification view. Table VI gives the pixel confusion matrix. On unseen data all stages are recall-heavy (recall 0.90–0.92) but over-segment: precision is only 0.66–0.67 against 0.81 for TG-LiteNet on DRIVE, and false positives (4.5–4.8% of the FOV) outnumber false negatives (0.8–1.0%) about five to one. The dominant error is spurious thin vessels and fragments, not missed vessels, which fits the Betti-0 counts; no stage changes the unseen precision by more than 0.014.

TABLE VI

PIXEL-LEVEL CONFUSION MATRIX AND CLASSIFICATION METRICS (VESSEL = POSITIVE CLASS, THRESHOLD 0.5). TP, FP, FN, TN ARE SHARES OF THE FOV PIXELS IN %; ALL VALUES ARE MEANS OVER IMAGES AND SEEDS. TOP: TG-LITENET (E) ON EVERY DATASET. BOTTOM: THE U-NET (A) AND TG-LITENET ON THE UNSEEN AVERAGE (OTHER STAGES: `RESULTS/TABLES/CONFUSION.CSV`). F1 EQUALS THE PER-IMAGE DICE.

	TP	FP	FN	TN	Precision	Recall	F1	Accuracy
DRIVE	10.91	2.59	2.14	84.37	0.809	0.839	0.822	0.953
STARE	9.65	3.91	0.82	85.62	0.712	0.921	0.801	0.953
CHASE_DB1	9.39	4.93	0.88	84.80	0.655	0.915	0.762	0.942
HRF	7.88	5.27	1.25	85.60	0.597	0.864	0.703	0.935
A, unseen avg.	8.94	4.45	1.02	85.59	0.669	0.897	0.762	0.945
E (det.), unseen avg.	8.97	4.70	0.98	85.34	0.655	0.900	0.755	0.943

Fragments, threshold and scale. Betti-0 error is dominated by tiny floating fragments: the ground truth has 3–12 components per image, the predictions 25–125. Table VII shows evaluation-only checks on seeds 3 and 4, with the cut-off and the threshold chosen on DRIVE-validation only. Removing components smaller than k px^2 at the 527-px FOV scale (k=100 for the U-Nets, 200 for the lite models) brings the unseen Betti-0 error of every stage to 4.7–5.1, including the plain U-Net (89.5 to 4.7), with Dice unchanged (within 0.002) and clDice within 0.005. The raw-count advantage of the topology loss is therefore not specific to it; what it keeps is the clDice gain. A validation-tuned threshold raises the unseen Dice of the topology-trained models (B +0.015, C +0.025, D +0.009, E +0.010) but lowers the plain U-Net's (−0.009), so part of the Dice cost of the topology loss at the fixed 0.5 threshold is miscalibration (recall 0.90 against precision 0.66); all main results keep the fixed threshold. Without the FOV rescaling the plain U-Net collapses (unseen Dice 0.768 to 0.637, HRF 0.50) while D and E hold (0.757 to 0.750, 0.756 to 0.751): the scale normalisation is essential for the U-Net, and since only A, D and E were run at native scale, on two seeds, we cannot say whether DG, the topology loss or the backbone gives the lite models this robustness.

B. Efficiency and the trade-off (RQ3)

TG-LiteNet has 147× fewer parameters and 70× fewer FLOPs than the U-Net, but is only 6.8× faster on four CPU threads in the Kaggle runs (429 vs 2915 ms), 7.5× on the T4 (12.2 vs 91.1 ms) and 8.8× in a controlled laptop benchmark (209 vs 1834 ms): depthwise convolutions have low arithmetic

TABLE VII

EVALUATION-ONLY CHECKS ON THE TRAINED MODELS (SEEDS 3 AND 4, UNSEEN AVERAGE). CLEANUP REMOVES CONNECTED COMPONENTS BELOW A SIZE CHOSEN ON DRIVE-VALIDATION; TUNED = ONE THRESHOLD PER MODEL CHOSEN ON DRIVE-VALIDATION; NATIVE = NO FOV RESCALING (RUN FOR A, D AND E ONLY).

	Betti-0 error		Dice			
Stage	fixed 0.5	cleanup	fixed 0.5	cleanup	tuned thr.	native
A	89.5	4.7	0.768	0.768	0.759	0.637
B	34.0	5.1	0.756	0.758	0.771	–
C	27.9	4.7	0.761	0.763	0.786	–
D	45.7	4.9	0.757	0.758	0.766	0.750
E (det.)	40.8	5.1	0.756	0.757	0.766	0.751

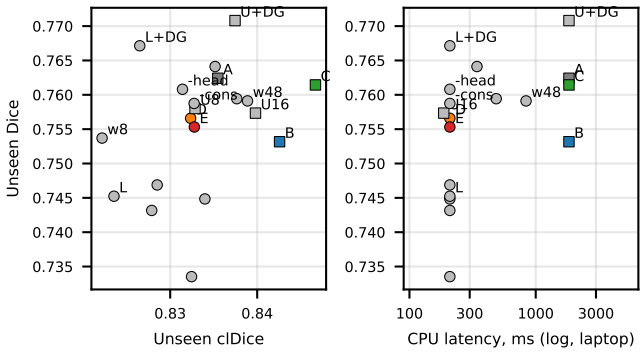


Fig. 3. Unseen Dice against unseen clDice (left) and CPU latency (right). Squares: U-Net, circles: lite; letters: stages A–E; grey: other configurations (Table X).

intensity, so FLOPs overstate the speed-up. Against plain compact U-Nets the picture reverses (Table X): a base-8 U-Net with the same FLOPs runs in 67 ms, 3.1× faster than the lite model, with the same Dice, and a base-16 U-Net (1.9 M parameters, within the 2 M budget) is as fast (187 ms) with better clDice and Betti-0. The depthwise-separable backbone therefore saves parameters, not time. Against the U-Net, TG-LiteNet gives up 0.008 DRIVE Dice, 0.006 unseen Dice and 0.003 unseen clDice; this loss is not larger on unseen domains (C–D Dice difference 0.004 on DRIVE, 0.005 unseen; identical gap 0.065). MC dropout multiplies the cost by T=20, so the uncertainty map is not free on edge hardware.

C. Uncertainty (RQ4)

Pixel-level uncertainty is moderately informative: ρ_{px} is 0.31 on DRIVE and slightly higher on the unseen domains (0.32 STARE, 0.34 CHASE_DB1, 0.34 HRF), so the expected increase under shift (RQ4) is present but small. Entropy is marginally better than MC variance at pixel level, variance at image level (ρ_{img} 0.151 vs 0.129 on DRIVE). Rejecting the 10% most uncertain pixels lowers the pixel error from 4.7% to 1.6% on DRIVE and from 6.5% to 3.5% on HRF. At image level the evidence is weak: AURC equals random on DRIVE and STARE, is better by 4% on CHASE_DB1 (ρ_{img}=0.48) and *worse* than random on HRF (ρ_{img}=−0.17). With 20–45 images per domain these statistics are imprecise, but they do not support referring whole images with MC variance, as expected under shift [26].

TABLE VIII

UNCERTAINTY QUALITY OF TG-LITENET (E). ρ_{px}: MEAN PER-IMAGE SPEARMAN CORRELATION BETWEEN PIXEL UNCERTAINTY AND MISCLASSIFICATION; ρ_{img}: SPEARMAN BETWEEN MEAN IMAGE UNCERTAINTY AND 1−DICE; AURC: AREA UNDER THE IMAGE-LEVEL RISK–COVERAGE CURVE (LOWER IS BETTER; RANDOM = MEAN ERROR, ORACLE = LOWER BOUND). *Var* = MC-DROPOUT VARIANCE, *Ent* = PREDICTIVE ENTROPY.

	ρ_{px}		ρ_{img}		AURC		
Domain	Var	Ent	Var	Ent	Var	Rand.	Oracle
DRIVE	0.311	0.313	0.151	0.129	0.178	0.178	0.167
STARE	0.318	0.325	0.085	-0.046	0.197	0.198	0.175
CHASE_DB1	0.341	0.343	0.477	0.270	0.227	0.237	0.210
HRF	0.340	0.352	-0.172	-0.386	0.305	0.297	0.252

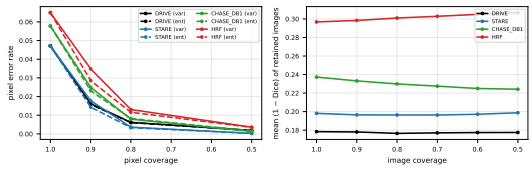


Fig. 4. Risk–coverage of TG-LiteNet on each domain: pixel error after rejecting the most uncertain pixels (left; solid = MC variance, dashed = entropy) and mean 1−Dice of the retained images when the most uncertain images are rejected (right).

D. Qualitative results and error analysis

In Fig. 5 (one image per dataset, seed 0, so illustrative only) the errors of all three models are mostly thin red (false positive) or blue (missed) fragments along the finest vessels; the large vessels are essentially error-free. On HRF the compact models miss more capillary-scale vessels than the U-Net (mean HRF sensitivity 0.864 for E vs 0.889 for A), which matches their lower HRF clDice. The uncertainty maps of E are faint and sparse on every domain; they follow some thin vessel segments and a few isolated spots, but by eye they do not single out the red and blue errors, consistent with the weak image-level triage above.

VII. ABLATION ANALYSIS

Table IX separates the factors (unseen averages; four cells have 3 seeds, four have 5). *Topology loss* lowers the raw Betti-0 error in all four pairs (U-Net 75→32 and 73→28, lite 86→46 and 88→45; every interval excludes zero, e.g. lite+DG -43 ± 9), raises clDice by 0.005–0.010 and changes unseen Dice by $+0.002$ to -0.010 (lite+DG: -0.011 ± 0.005). It therefore still helps a ≤2 M-parameter network in raw Betti-0 and clDice, with the cleanup caveat above. *DG augmentation* adds 0.008 to 0.022 unseen Dice, narrows the gap (0.068→0.060 for the plain U-Net) and leaves Betti-0 unchanged without a topology loss. *Backbone*: the compact network loses 0.004–0.017 Dice and 0.010–0.015 clDice and adds 11–17 raw Betti-0 error in every cell. The highest unseen Dice come from DG without topology loss (U-Net 0.771; lite 0.767 with 0.21 M parameters, $+0.005 \pm 0.008$ over the baseline), but that lite model has worse clDice (-0.009 ± 0.003) than the

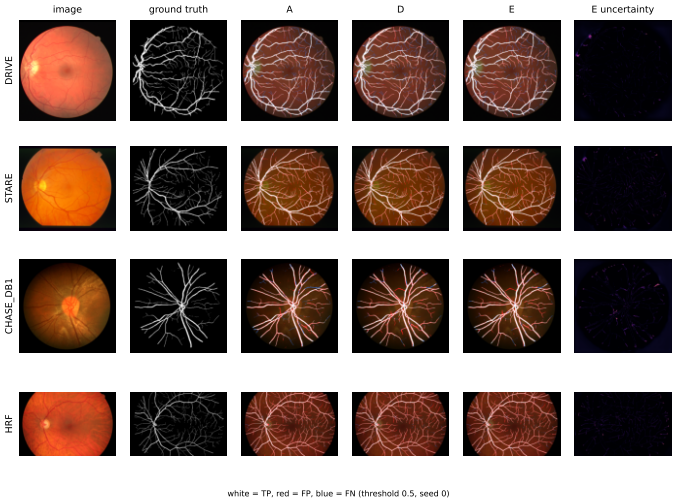


Fig. 5. Qualitative results (seed 0): image, ground truth, error maps of stages A, D and E, and the MC-dropout uncertainty of E.

TABLE IX
FULL FACTORIAL DESIGN (BACKBONE × TOPOLOGY × DG). UNSEEN = MACRO AVERAGE OVER THE THREE TARGETS. U-NET PLAIN, U-NET+TOPO, U-NET+TOPO+DG AND LITE+TOPO+DG ARE STAGES A–D (3 SEEDS); THE OTHER CELLS HAVE 2 SEEDS, SO THEIR ± IS ONLY INDICATIVE.

Backbone	Topo	DG	Src Dice	Unseen Dice	Unseen clDice	ΔDice
U-Net	–	–	0.830±0.001	0.762±0.006	0.835±0.002	0.068±0.006
U-Net	–	✓	0.831±0.000	0.771±0.001	0.837±0.000	0.060±0.001
U-Net	✓	–	0.827±0.001	0.753±0.004	0.843±0.003	0.074±0.004
U-Net	✓	✓	0.826±0.001	0.761±0.001	0.847±0.001	0.065±0.001
Lite	–	–	0.823±0.001	0.745±0.011	0.824±0.001	0.078±0.011
Lite	–	✓	0.824±0.000	0.767±0.002	0.827±0.001	0.057±0.002
Lite	✓	–	0.821±0.002	0.747±0.007	0.829±0.004	0.074±0.009
Lite	✓	✓	0.822±0.001	0.757±0.003	0.832±0.002	0.065±0.003

TABLE X
VARIANTS OF STAGE D: TOPOLOGY LOSS (TOP), REMOVED INGREDIENTS AND PLAIN COMPACT U-NETS WITH THE SAME RECIPE (MIDDLE), LITE WIDTH (BOTTOM). STAGE D HAS 5 SEEDS, ALL OTHER ROWS 3. CPU LATENCY (4 THREADS) IS FROM A CONTROLLED BENCHMARK ON ONE LAPTOP CPU (SEC. V).

	Src Dice	Unseen Dice	Unseen clDice	Unseen β_0	Params (M)	GFLOPs	CPU (ms)
soft-clDice + skel. head (D)	0.822±0.001	0.757±0.003	0.832±0.002	44.7	0.21	5.5	209
cbDice	0.818±0.001	0.745±0.002	0.834±0.002	39.1	0.21	5.5	209
Skeleton Recall	0.814±0.001	0.743±0.006	0.828±0.001	55.3	0.21	5.5	209
clDice + Skel. Recall	0.810±0.001	0.734±0.001	0.832±0.003	36.5	0.21	5.5	209
D without skeleton head	0.822±0.000	0.761±0.006	0.831±0.001	53.8	0.21	5.5	209
D without consistency loss	0.822±0.001	0.759±0.005	0.833±0.002	52.2	0.21	5.5	209
plain U-Net, base 8 (topo+DG)	0.821±0.000	0.758±0.006	0.833±0.001	48.9	0.49	6.0	67
plain U-Net, base 16 (topo+DG)	0.823±0.001	0.757±0.005	0.840±0.001	33.7	1.94	24.1	187
width w=8	0.816±0.000	0.754±0.005	0.822±0.001	61.1	0.06	1.6	76
width w=16 (D)	0.822±0.001	0.757±0.003	0.832±0.002	44.7	0.21	5.5	209
width w=24	0.822±0.003	0.764±0.002	0.835±0.003	40.0	0.46	11.8	343
width w=32	0.823±0.001	0.759±0.005	0.838±0.003	34.3	0.81	20.3	487
width w=48	0.823±0.002	0.759±0.005	0.839±0.002	30.1	1.81	44.4	838

baseline. Which configuration is best depends on whether overlap or connectivity matters more; the full TG-LiteNet recipe is a connectivity-oriented operating point. In Table X (top) the default soft-clDice has the best Dice. cbDice and clDice+Skeleton Recall lower the raw Betti-0 error a little (44.7 to 39.1 and 36.5; within the seed spread) but cost 0.012 (±0.005) and 0.023 (±0.004) unseen Dice, and Skeleton Recall alone is worse in clDice (−0.004, ±0.003), so none beats soft-clDice. Removing single ingredients (middle) shows that neither the skeleton head nor the consistency loss helps: without the head, unseen Dice is 0.761 and clDice 0.831 (stage D: 0.757 and 0.832); without the consistency term 0.759 and 0.833; the raw Betti-0 error is higher by 9 (±8) without the head and by 7 (±14) without the consistency term, the only trace of a benefit, and raw counts are not a robust measure given the cleanup check. The plain compact U-Nets with the stage-D recipe match the lite model in Dice (base 8, 0.49 M parameters: 0.758; base 16, 1.9 M: 0.757), and the larger one is more connected (clDice 0.840 vs 0.832, Betti-0 33.7 vs 44.7). In the width sweep (bottom), unseen Dice is 0.754 at w=8, 0.757 at 16, 0.764 at 24 and 0.759 at 32 and 48, whereas clDice (0.822 to 0.839) and Betti-0 (61 to 30) keep improving and latency grows from 76 to 838 ms. Capacity stops paying off for overlap near w=24 while topology improves up to w=48.

A. *Limitations and threats to validity*

(i) Only 16 training images; seed-to-seed variation can match small gains (stage A's unseen Dice ranges from 0.755 to 0.769 over its five seeds). Stages A–E use 5 seeds, all other configurations 3 and the evaluation-only checks 2. (ii) The unseen average weights three very different datasets equally; raw Betti counts grow with image size and are dominated by HRF. (iii) The no-rescale evaluation covers only A, D and E on two seeds. (iv) Labels are the first observer only and four of the five FOV sets are estimated. (v) Soft-clDice, Skeleton Recall and cbDice [8] are our re-implementations. (vi) Latency is eager-PyTorch, framework-level, not an edge measurement; identical models varied up to 2× between runs on shared cloud CPUs, so architectures are compared with a controlled laptop benchmark. (vii) The target datasets were never used for fitting or checkpoint selection, but we inspected their results while designing the ablations, so they are held-out rather than untouched, and the many comparisons are uncorrected; the cleanup size and the per-model thresholds were chosen on four DRIVE-validation images and are post-hoc analyses. (viii) The DG stage bundles augmentation, a

consistency term and a second view per step; the controls remove only the consistency term and the skeleton head. (ix) Source-only evaluation is harder than multi-source or test-time adaptation, so our absolute numbers are not comparable with such papers [15], [16], [22].

VIII. Conclusion and Future Work

We built and evaluated TG-LiteNet, a compact, source-only-trained vessel segmenter, with a staged ablation, a factorial design, capacity, loss and ingredient variants, plain compact U-Net baselines, evaluation-only controls and an uncertainty analysis on three held-out datasets. (O1) The pipeline is source-only and reproducible; every number comes from 70 trained models. (O2) The topology loss lowers the raw unseen Betti-0 error of the U-Net from 75 to 32 and raises clDice by 0.007 at 0.009 Dice, but simple cleanup brings every model to Betti-0 $\approx$5, so its lasting benefit is the clDice gain (RQ1); DG augmentation adds 0.008 unseen Dice and narrows the shift gap from 0.074 to 0.065 (RQ2); the compact backbone costs 0.005 Dice, 0.014 clDice and 17 raw Betti-0 (RQ3). TG-LiteNet is 0.006 Dice below the baseline U-Net on unseen data (0.756 vs 0.762) with 147$\times$ fewer parameters and 70$\times$ fewer FLOPs, but plain compact U-Nets with the same recipe match its accuracy (base 8: 0.758) or its connectivity and more (base 16: clDice 0.840) at equal or lower CPU latency, so the depthwise backbone saves parameters, not time. Removing the skeleton head or the consistency loss lowers neither Dice nor clDice (the head may trim raw fragments slightly) and dropout adds nothing measurable; the Dice-optimal configuration is DG without topology loss. (O3) MC-dropout ranks pixel errors moderately (ρ_{px} 0.31–0.34), but image-level rejection is within 4% of random and worse on HRF, so RQ4 is only partly answered. **Limitations** are listed in Sec. VII-A. **Future work:** the official cbDice code, a no-rescale evaluation of all stages, held-out FIVES and multi-source DG, test-time adaptation as an upper bound, INT8 quantisation on a real edge device, and calibrated uncertainty before any referral decision.

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